INNOVATE BHARAT HACKATHON 2026

TRACKS:

- AI & Intelligent Systems (AIIS)
- Web app and software Innovation (WASI)
- Cybersecurity and Block Chain (CSBC)
- Data Science and Smart Analysis (DSSA)
- Social Impact and Smart India Solutions (SISIS)
- School Student Innovation (SCHII)

AI-Powered Virtual Interviewer for Adaptive Mock Interviews

Team Name: Volfe
Team ID: AIIS188

Team Members:
1.Muhammad Shayaan Ali
2.Arnav Neel Thakur
3.Vakar Younish
4.Aditya Kumar Jha
5.Noel Anto Shaji
6.Swayam Jaswal

Problem Statement

- Many candidates fail interviews due to:
  - Lack of real interview practice
  - Poor communication skills
  - Interview anxiety
- Existing platforms:
  - Are static or chatbot-based
  - Do not simulate real interview environments
  - Lack personalized feedback
- There is a need for an **AI-driven, adaptive, and realistic mock interview system**

Proposed Solution/Idea

- Develop an **AI-powered virtual interviewer**
- Uses a **human-like AI avatar** for realistic interaction
- Conducts:
  - HR interviews
  - Technical interviews
- Adapts questions based on:
  - Role
  - Company
  - Experience level
- Provides **real-time evaluation and feedback**

Innovation & Uniqueness

- 🎯 Adaptive question generation (not static questions)
- 🤖 Human-like AI avatar (face-to-face simulation)
- 🧠 Real-time AI-based evaluation
- 📊 Selection probability prediction
- 🎙️ Voice-based interaction (Speech-to-Text & Text-to-Speech)
- 📈 Personalized feedback system

Technology Used

Programming: Python, JavaScript

AI/ML:

- Natural Language Processing (NLP)
- Machine Learning models

Speech Processing:

- Speech-to-Text (STT)
- Text-to-Speech (TTS)

Frontend: React.js

Backend: Django / Node.js

Optional: Facial Expression Analysis

Features & Functionalities

- 👤 AI Avatar conducts interview
- 🎯 Role-based question generation
- 🎙️ Voice & text interaction
- 🧠 Real-time answer evaluation
- 📊 Performance scoring system
- 📄 Detailed feedback report
- 📈 Selection probability estimation

Implementation Plan / Roadmap

Phase 1: Research & Problem Analysis

Phase 2: System Design & Architecture

Phase 3: AI Model Development

Phase 4: Voice Interaction Integration

Phase 5: Frontend & Avatar Integration

Phase 6: Testing & Evaluation

Phase 7: Final Deployment